

Announced Rescue Pricing with Latent Market Thickness

Group-meeting note — exact benchmark, formal setup, and rescue value across thickness

26 August 2026

Main result. Write $V(m)$ for the optimized completion gain of the best rescue menu over the best flat price at public thickness m . We first solve the cutoff-WPBE and optimal menu for a publicly known single driver, then embed the same mechanism in latent Poisson supply. In the uniform no-entry model with $\beta \geq 1/2$, $V(m) > 0$ at every finite thickness and converges to zero at both thickness endpoints. When $\alpha = 1$ and $\gamma = 0$, the exact-one-driver gain is the coefficient of m in the thin-market expansion of $V(m)$.

1 Exact one-rider/one-driver benchmark and optimal menu

1.1 Closed-form solution

Here the number of incumbents is publicly known to be exactly one. The rider has $v \sim U[0, 1]$, the driver has $c \sim U[0, 1]$, and the platform maximizes the probability of completion. The rider's delayed value is βv ; the incumbent's delayed surplus is $\delta(p_j - c)$. The platform commits to $0 \leq p_1 \leq p_2 \leq 1$, with $\beta, \delta \in (0, 1]$; $j \in \{1, 2\}$ indexes the active payment. There is no fresh entry in this benchmark.

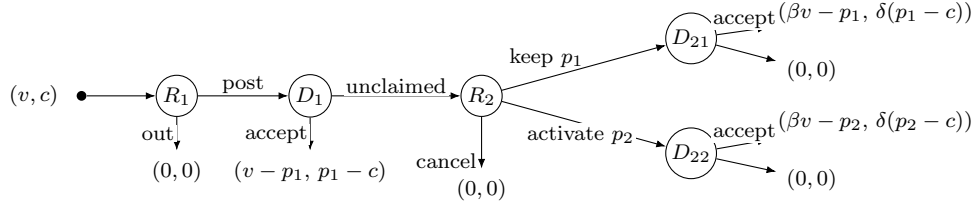


Figure 1: Extensive form.

For an interior active menu $0 \leq p_1 < p_2 < \beta$, consider an anonymous pure-cutoff WPBE in which the driver accepts now iff $c \leq \hat{c}$ and both continuation actions occur for positive rider masses. In equilibrium the rider posts exactly when $v \geq p_1$. Bayes' rule gives $v \mid \text{post} \sim U[p_1, 1]$ and, following refusal, $c \mid \text{refusal} \sim U[\hat{c}, 1]$. Following refusal, the probability that the driver accepts p_j is $(p_j - \hat{c})/(1 - \hat{c})$. Equating the rider's expected payoffs from repeating p_1 and activating p_2 gives the switching value $v^M(\hat{c})$ and the regions

$$\begin{aligned} v < p_1/\beta & : \text{cancel,} \\ p_1/\beta \leq v \leq v^M(\hat{c}) & : \text{repeat } p_1, \\ v > v^M(\hat{c}) & : \text{activate rescue } p_2, \end{aligned} \quad v^M(\hat{c}) = \frac{p_1 + p_2 - \hat{c}}{\beta}.$$

Conditional on a posted request, the rider masses who repeat and rescue are $(p_2 - \hat{c})/[\beta(1 - p_1)]$ and $(\beta - p_1 - p_2 + \hat{c})/[\beta(1 - p_1)]$. Hence the cutoff driver's discounted wait payoff is

$$\begin{aligned} W_1(\hat{c}) &= \frac{\delta}{\beta(1 - p_1)} \{ (p_2 - \hat{c})(p_1 - \hat{c}) + (\beta - p_1 - p_2 + \hat{c})(p_2 - \hat{c}) \} \\ &= \frac{\delta(\beta - p_2)(p_2 - \hat{c})}{\beta(1 - p_1)}. \end{aligned}$$

Equating $W_1(\hat{c})$ to the immediate payoff $p_1 - \hat{c}$ and solving for the cutoff gives

$$\hat{c}(p_1, p_2) = \left[p_1 - \frac{\delta(p_2 - p_1)(\beta - p_2)}{\beta(1 - p_1) - \delta(\beta - p_2)} \right]_+. \quad (1)$$

where $[x]_+ = \max\{x, 0\}$ implements the reject-all boundary. At $p_2 = p_1$, $p_2 \geq \beta$, or $p_1 \geq \beta$, the rescue branch is inactive and the cutoff is $\hat{c} = p_1$.

Immediate completion has probability $(1 - p_1)\widehat{c}$. The unconditional repeat and rescue completion probabilities are, respectively,

$$M_{\text{rep}} = \frac{(p_2 - \widehat{c})(p_1 - \widehat{c})}{\beta}, \quad M_{\text{res}} = \frac{(\beta - p_1 - p_2 + \widehat{c})(p_2 - \widehat{c})}{\beta}.$$

Their sum collapses to the second term below, so total completion is

$$M_1(p_1, p_2) = (1 - p_1)\widehat{c} + \frac{(\beta - p_2)(p_2 - \widehat{c})}{\beta}. \quad (2)$$

If $\widehat{c} > 0$, the same marginal-driver indifference can be written as

$$(1 - p_1)(p_1 - \widehat{c}) = \frac{\delta}{\beta}(\beta - p_2)(p_2 - \widehat{c}), \quad (3)$$

and therefore

$$M_1 = (1 - p_1) \left[\widehat{c} + \frac{p_1 - \widehat{c}}{\delta} \right] = p_1(1 - p_1) + \frac{(1 - p_1)(1 - \delta)}{\delta}(p_1 - \widehat{c}). \quad (4)$$

Here $p_1(1 - p_1)$ is completion under a flat menu at the same first price; the last term is therefore the within- p_1 gain created by rescue. For $0 < \widehat{c} < p_1$, a rescue menu strictly improves completion relative to the same p_1 when $\delta < 1$. Every positive-cutoff active menu is neutral relative to the same p_1 when $\delta = 1$.

One-dimensional reduction. Fix a target cutoff $\widehat{c}$. The continuation component depending on p_2 is $H(p_2; \widehat{c}) = (\beta - p_2)(p_2 - \widehat{c})/\beta$, with

$$H_{p_2} = \frac{\beta + \widehat{c} - 2p_2}{\beta}, \quad H_{p_2 p_2} = -\frac{2}{\beta} < 0.$$

Thus its unique envelope maximizer is

$$\widehat{p}_2(\widehat{c}) = \frac{\beta + \widehat{c}}{2}.$$

At this rescue price, (3) becomes

$$(1 - p_1)(p_1 - \widehat{c}) = \frac{\delta(\beta - \widehat{c})^2}{4\beta}.$$

The two quadratic roots are $[1 + \widehat{c} \pm \sqrt{\Delta(\widehat{c})}]/2$. The plus root exceeds $(1 + \widehat{c})/2 \geq \widehat{p}_2(\widehat{c})$ and violates $p_1 \leq p_2$; hence the feasible root is

$$\Delta(\widehat{c}) = (1 - \widehat{c})^2 - \frac{\delta}{\beta}(\beta - \widehat{c})^2, \quad \widehat{p}_1(\widehat{c}) = \frac{1 + \widehat{c} - \sqrt{\Delta(\widehat{c})}}{2}. \quad (5)$$

The best flat menu solves $\max_{p_1 \in [0, 1]} p_1(1 - p_1) = 1/4$. Substituting the two envelope prices into (2) gives $J_\delta(\widehat{c})$, so the platform compares the best active menu with this flat benchmark:

$$M_1^* = \max \left\{ \frac{1}{4}, \max_{\widehat{c} \in [0, \beta]} J_\delta(\widehat{c}) \right\}, \quad J_\delta(\widehat{c}) = \frac{\widehat{c}[1 - \widehat{c} + \sqrt{\Delta(\widehat{c})}]}{2} + \frac{(\beta - \widehat{c})^2}{4\beta}. \quad (6)$$

Optimal menu. If $\beta \leq 1/2$, a flat menu $p_1 = p_2 = 1/2$ is optimal and $M_1^* = 1/4$. If $\delta = 1$, the same value obtains for every β . If $\beta > 1/2$ and $\delta < 1$, J_δ is strictly concave and has a unique interior maximizer $\hat{c}^* \in (0, \beta)$; the optimal rescue menu is $p_1^* = \hat{p}_1(\hat{c}^*) < p_2^* = \hat{p}_2(\hat{c}^*)$.

For $\beta > 1/2$ and $\delta < 1$, the unique interior cutoff solves

$$1 - \delta\beta + 3(\delta - 1)\hat{c} + 2(1 - \delta/\beta)\hat{c}^2 = \frac{2\beta - 1}{\beta}\hat{c}\sqrt{\Delta(\hat{c})}. \quad (7)$$

Indeed,

$$\Delta'(\hat{c}) = 2(\delta - 1) + 2(1 - \delta/\beta)\hat{c}, \quad J'_\delta(\hat{c}) = \frac{1}{2} \left[\sqrt{\Delta} + \hat{c}(1/\beta - 2) + \frac{\hat{c}\Delta'}{2\sqrt{\Delta}} \right],$$

and clearing the positive denominator in $J'_\delta(\hat{c}) = 0$ gives (7).

1.2 Numerical illustration

For the calibration used below, $\beta = \delta = 4/5$. Equation (7) becomes

$$12 - 20\hat{c} = 5\hat{c}\sqrt{9 - 10\hat{c}}.$$

Squaring gives $(5\hat{c} - 4)(50\hat{c}^2 + 75\hat{c} - 36) = 0$. The root $\hat{c} = 4/5$ violates the unsquared equation, while the negative quadratic root is infeasible. The remaining quadratic has discriminant $75^2 + 4(50)(36) = 225 \cdot 57$. Therefore

$$\hat{c}^* = \frac{3(\sqrt{57} - 5)}{20}, \quad p_1^* = \frac{11 + \sqrt{57}}{40}, \quad p_2^* = \frac{1 + 3\sqrt{57}}{40}, \quad (8)$$

Substituting $\hat{c}^*$ into (5) gives the two prices; substituting the resulting menu into (2) gives

$$M_1^* = \frac{1261 - 57\sqrt{57}}{3200} = 25.9581\%, \quad 100 \left(M_1^* - \frac{1}{4} \right) = 0.9581 \text{ percentage points.}$$

Unconditional completion decomposes into 20.5104% in the first window, 2.1208% after repeating p_1 , and 3.3269% after activating p_2 . These components sum to $M_1^* = 25.9581\%$. The menu and decomposition depend on (β, δ) .

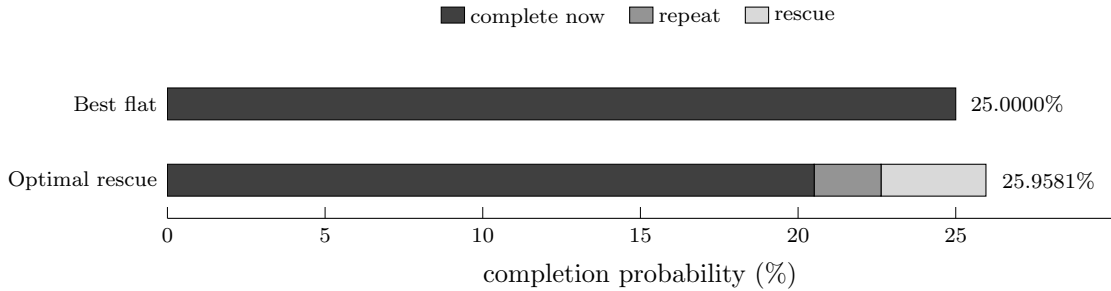


Figure 2: Exact-one-driver illustration at $\beta = \delta = .8$. The contingent menu adds 0.9581 percentage points over the best flat price; its completion probability is decomposed by the period and payment used.

Type regions across (β, δ) . For every active calibration below, the unique cutoff is the sign-screened root of (7); prices then come from (5). Because v and c are uniform, interval widths are also population shares. In the order used in Figure 3,

$$\begin{aligned} \text{rider widths: } & \left(p_1, \frac{p_1}{\beta} - p_1, \frac{p_2 - \hat{c}}{\beta}, \frac{\beta - p_1 - p_2 + \hat{c}}{\beta} \right), \\ \text{driver widths: } & (\hat{c}, p_1 - \hat{c}, p_2 - p_1, 1 - p_2). \end{aligned}$$

The four rider entries mean no post, cancel after failure, repeat p_1 , and activate p_2 . The four driver entries mean accept now, wait and accept either terminal price, accept only the rescue price, and never accept. The optimal menus used in the plot are

β	δ	$\hat{c}^*$	p_1^*	p_2^*	M_1^*
.6	.5	.460069	.467733	.530034	.253038
.6	.8	.478613	.488210	.539306	.251089
.8	.5	.370421	.420148	.585210	.272458
.8	.8	.382475	.463746	.591238	.259581
.9	.5	.329857	.405843	.614928	.286282
.9	.8	.314018	.453693	.607009	.266932

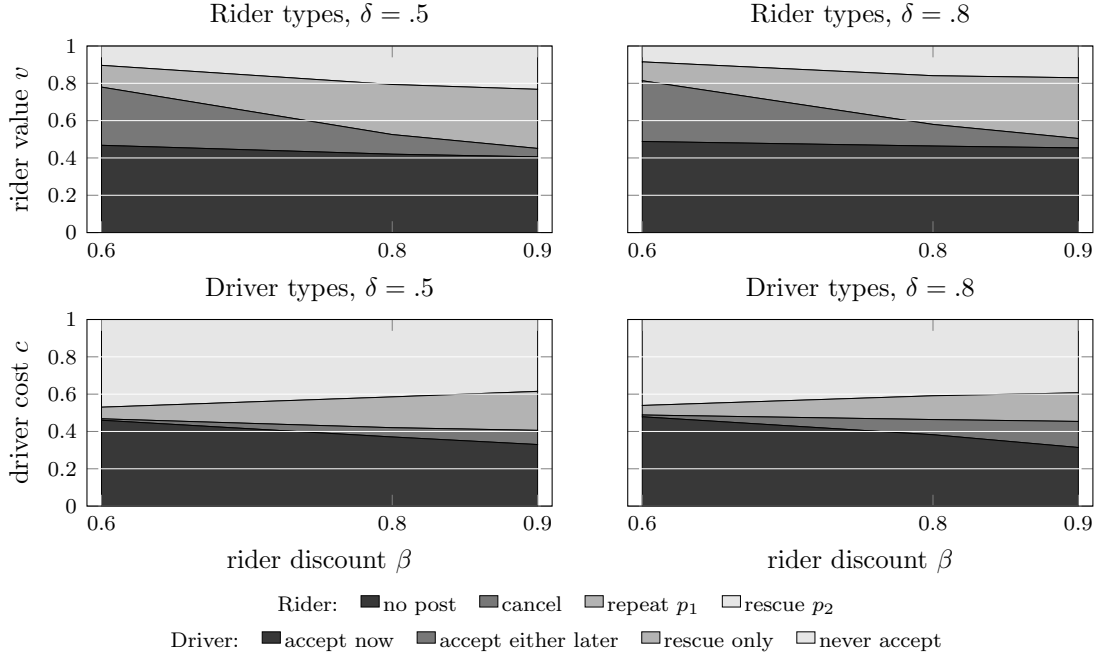


Figure 3: Equilibrium type maps for six optimal active menus. A vertical slice partitions the full type interval $[0, 1]$; hence its shaded heights are the uniform-population shares reported above. The vertices are the recomputed calibrations at $\beta = .6, .8, .9$ and the connecting segments are visual guides, not a continuous comparative-statics claim. Continuation labels describe actions conditional on initial failure, not realized path probabilities.

When $\beta \leq 1/2$, the optimal policy is the flat menu $p_1 = p_2 = 1/2$ and the active-rescue regions collapse. At $\delta = 1$, a flat menu is a canonical optimizer; continuation actions after its rejection are payoff-irrelevant.

2 Formal market-thickness setup

Market thickness is public; the realized supply count is latent and individual types are private. The platform commits to $0 \leq p_1 \leq p_2 \leq 1$. The primitives are:

Primitive	Domain	Definition
m	$(0, \infty)$	Public mean of incumbent supply; the latent count is $N^I \sim \text{Pois}(m)$.
α	$[0, 1]$	Probability that a waiting incumbent remains available in the terminal window.
γ	$[0, \infty)$	Fresh-entry intensity relative to m ; after failure, $N^E \sim \text{Pois}(\gamma m)$.
β	$(0, 1]$	Rider discount factor; delayed gross value is βv .
δ	$(0, 1]$	Incumbent discount factor; delayed surplus at p_j is $\delta(p_j - c_i)$.
F	continuous CDF on $[0, 1]$	Distribution of driver costs $c_i \sim F$; draws are i.i.d. and an incumbent's cost persists across windows.
G	continuous CDF on $[0, 1]$	Distribution of the rider's private value $v \sim G$.

All primitives and the menu are common knowledge; realizations are observed only as specified below. The numerical analysis sets $F = G = U[0, 1]$ and $\gamma = 0$. In $V(m)$, only m varies; $(\alpha, \gamma, \beta, \delta, F, G)$ are held fixed. Throughout, $j \in \{1, 2\}$ indexes the payment used in the terminal window.

2.1 Timing and information

Stage 1. Commitment. The platform observes $m > 0$ and commits publicly to $0 \leq p_1 \leq p_2 \leq 1$. It cannot revise either payment after subsequent actions, and p_2 can be activated only after first-round universal rejection.

Stage 2. Rider entry. The rider privately observes $v \sim G$, observes (m, p_1, p_2) , and chooses whether to post. Leaving gives zero; posting itself generates no transfer.

Stage 3. Latent incumbent supply. Conditional on a posted request,

$$N^I \sim \text{Pois}(m).$$

The realization is latent to the platform, rider, and drivers. Each incumbent privately observes only her persistent cost $c_i \sim F$ and simultaneously chooses whether to accept p_1 or wait. She does not observe the number, costs, or actions of rival drivers when deciding.

Stage 4. Initial assignment or public failure. If at least one incumbent accepts, one acceptor is selected uniformly and the request completes with selected-pair payoff $(v - p_1, p_1 - c_i)$; unselected acceptors receive zero. If none accepts, public failure reveals only universal rejection, not the realized count or costs.

Stage 5. Rider continuation. After failure, the rider chooses cancel, repeat p_1 , or activate p_2 using v , the committed menu, and the failure posterior.

Stage 6. Survival, entry, and terminal assignment. Following continuation, each waiting incumbent remains available independently with probability α . Independently, fresh supply $N^E \sim \text{Pois}(\gamma m)$ arrives and draws i.i.d. costs from F . Counts, survival realizations, and rival costs remain latent. At the active payment p_j , available drivers make their terminal accept decisions. If at least one accepts, one is selected uniformly. The rider receives $\beta v - p_j$; a selected incumbent receives $\delta(p_j - c_i)$, while a selected fresh entrant receives $p_j - c_i$. If nobody accepts, the request expires and all players receive zero.

Benchmark distinction. Unlike the benchmark above, $m = 1$ denotes a hidden $N^I \sim \text{Pois}(1)$ count, not a publicly known single driver.

Equilibrium class. We study anonymous symmetric pure-cutoff WPBE. An incumbent accepts p_1 exactly when c_i lies below a cutoff $\hat{c}$. Beliefs follow Bayes' rule after posting and universal rejection. Poisson splitting determines continuation supply after failure, while Palm conditioning leaves a focal incumbent's number of rivals Poisson with mean m . Thus $\hat{c}$ is derived from accept-versus-wait indifference, not selected from an observed driver count. The formal setup allows $\gamma \geq 0$; the global optimization and numerical evidence below specialize to $\gamma = 0$.

WPBE equations and completion

For a candidate accept-now cutoff $\hat{c}$, define

$$\phi(x) = \frac{1 - e^{-x}}{x}, \quad \phi(0) = 1, \quad \lambda_j(\hat{c}) = m\{\alpha[F(p_j) - F(\hat{c})] + \gamma F(p_j)\}, \quad C_j(\hat{c}) = 1 - e^{-\lambda_j(\hat{c})}.$$

For $K \sim \text{Pois}(x)$,

$$\Pr(K = 0) = e^{-x}, \quad \mathbb{E}\left[\frac{1}{1 + K}\right] = \frac{1 - e^{-x}}{x} = \phi(x).$$

Thus $\phi(x)$ is a focal acceptor's expected assignment share when her rivals have Poisson mean x . Poisson thinning gives $\lambda_j(\hat{c})$ as the mean terminal supply eligible at p_j , and $C_j(\hat{c})$ is the probability that this eligible set is nonempty. After universal rejection the rider compares

$$0, \quad C_1(\hat{c})(\beta v - p_1), \quad C_2(\hat{c})(\beta v - p_2).$$

If $C_2(\hat{c}) > C_1(\hat{c})$, the repeat/rescue boundary is

$$v^M(\hat{c}) = \frac{p_2 C_2(\hat{c}) - p_1 C_1(\hat{c})}{\beta[C_2(\hat{c}) - C_1(\hat{c})]}. \quad (9)$$

Equation (9) comes from equating the last two payoffs. If $C_2(\hat{c}) = C_1(\hat{c})$, rescue adds no coverage and we set $v^M(\hat{c}) = +\infty$. Conditional on posting, define

$$\eta^{\text{rep}}(\hat{c}) = \Pr\{p_1/\beta \leq v \leq v^M(\hat{c}) \mid v \geq p_1\}, \quad \eta^{\text{res}}(\hat{c}) = \Pr\{v > v^M(\hat{c}) \mid v \geq p_1\},$$

with empty regions assigned zero mass. A focal incumbent of type c compares

$$\begin{aligned} A(c; \hat{c}) &= \phi(mF(\hat{c}))(p_1 - c), \\ W(c; \hat{c}) &= \delta \alpha e^{-mF(\hat{c})} \{\eta^{\text{rep}}(\hat{c}) \phi(\lambda_1(\hat{c}))(p_1 - c)^+ + \eta^{\text{res}}(\hat{c}) \phi(\lambda_2(\hat{c}))(p_2 - c)^+\}. \end{aligned} \quad (10)$$

Here $x^+ = \max\{x, 0\}$. Both payoffs are conditional on a posted request. In A , $\phi(mF(\hat{c}))$ is the focal driver's initial assignment share. In W , $e^{-mF(\hat{c})}$ is the probability that all rivals reject initially, α is the focal incumbent's survival probability, and $\phi(\lambda_j)$ is her terminal assignment share. An interior cutoff satisfies $A(\hat{c}; \hat{c}) = W(\hat{c}; \hat{c})$; one-sided inequalities apply at 0 and p_1 . Completion is

$$M(p_1, p_2; \hat{c}) = (1 - G(p_1)) \left[1 - e^{-mF(\hat{c})} + e^{-mF(\hat{c})} \{\eta^{\text{rep}}(\hat{c}) C_1(\hat{c}) + \eta^{\text{res}}(\hat{c}) C_2(\hat{c})\} \right]. \quad (11)$$

The factor $1 - G(p_1)$ is the posting probability. Inside the brackets, the first term is initial coverage and the second is failure followed by terminal recovery.

Selection. In the uniform no-entry model, every menu has a unique *equilibrium cutoff value* within this WPBE class; null-type actions and payoff-irrelevant beliefs may differ.

Literature and contribution. The solve-threshold-then-optimize architecture follows Aviv and Pazgal (2008), and contingent preannouncement follows Correa et al. (2016). Waiting costs and successive offers connect to Carare and Rothkopf (2005) and Chen (2012). Discrete Dutch clocks, population uncertainty, and hidden group size connect to Li and Kuo (2013), Myerson (1998), and Buchanan et al. (2016); two-sided platform timing connects to Hu et al. (2022). The contribution is not any one primitive. It is the cutoff-WPBE and completion design that jointly handle latent same-request competition, discounted incumbent delay, and the rider’s private continuation decision.

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3 Solving $p_1^*(m), p_2^*(m)$ from the WPBE

Set $F = G = U[0, 1]$, $\gamma = 0$, $\alpha \in (0, 1]$, and $m > 0$. Cutoff indifference yields the following scalar solution.

Step 1: solve the rescue-price envelope at a fixed cutoff. Let $k = \alpha m$, the effective mean surviving-incumbent supply per unit cost mass. Under uniform types and no entry, $C_j = 1 - e^{-k(p_j - \hat{c})}$. Substituting the switching value (9) makes repeat and rescue completion telescope:

$$C_1 \left(v^M - \frac{p_1}{\beta} \right) + C_2(1 - v^M) = C_2 \left(1 - \frac{p_2}{\beta} \right).$$

Therefore the continuation term, before multiplication by the first-round failure probability $e^{-m\hat{c}}$, is

$$\frac{S_m(\hat{c}, p_2)}{\beta}, \quad S_m(\hat{c}, p_2) = (\beta - p_2)\{1 - e^{-k(p_2 - \hat{c})}\}.$$

For fixed $\hat{c}$, this expression is strictly concave in $p_2 \in [\hat{c}, \beta]$. Indeed,

$$\frac{\partial S_m}{\partial p_2} = -(1 - e^{-k(p_2 - \hat{c})}) + k(\beta - p_2)e^{-k(p_2 - \hat{c})}, \quad \frac{\partial^2 S_m}{\partial p_2^2} = -e^{-k(p_2 - \hat{c})}\{2k + k^2(\beta - p_2)\} < 0.$$

The first-order condition for its unique maximizer $\hat{p}_2(\hat{c}; m)$ is

$$e^{k(\hat{p}_2 - \hat{c})} - 1 = k(\beta - \hat{p}_2). \tag{12}$$

Let $z = k(\beta - \hat{p}_2)$ and $y = 1 + z$. Equation (12) gives $ye^y = e^{1+k(\beta - \hat{c})}$, so

$$\hat{p}_2(\hat{c}; m) = \beta - \frac{W_0(e^{1+k(\beta - \hat{c})}) - 1}{k}. \tag{13}$$

Here W_0 is the principal Lambert- W function, defined by $W_0(x)e^{W_0(x)} = x$. Its argument is positive, and W_0 is the unique real branch that delivers $z \geq 0$ and $\hat{c} < \hat{p}_2 < \beta$ for an interior cutoff. Since $e^x - 1 > x$ for $x > 0$, (12) also implies $\hat{p}_2 < (\beta + \hat{c})/2$. Write $S_m(\hat{c}) = S_m(\hat{c}, \hat{p}_2(\hat{c}; m))$.

Step 2: impose the driver's WPBE indifference. Define

$$h_m(\hat{c}) = \frac{e^{m\hat{c}} - 1}{\hat{c}}, \quad h_m(0) = m. \quad (14)$$

The value at zero follows from continuity. The function h_m rewrites the cutoff indifference in terms of the odds of reaching the failure node. The direct cutoff equation is

$$(p_1 - \hat{c})h_m(\hat{c}) = \frac{\delta S_m(\hat{c})}{\beta(1 - p_1)}. \quad (15)$$

Multiplying by $1 - p_1$ and writing $R_m(\hat{c}) = \delta S_m(\hat{c})/[\beta h_m(\hat{c})]$ gives the quadratic

$$(1 - p_1)(p_1 - \hat{c}) = R_m(\hat{c}), \quad p_1^2 - (1 + \hat{c})p_1 + \hat{c} + R_m(\hat{c}) = 0.$$

Its roots are $[1 + \hat{c} \pm \sqrt{(1 - \hat{c})^2 - 4R_m(\hat{c})}]/2$. The plus root exceeds $(1 + \hat{c})/2 > \hat{p}_2$ and violates $p_1 \leq p_2$; the implementing root is

$$\hat{p}_1^\delta(\hat{c}; m) = \frac{1 + \hat{c} - \sqrt{(1 - \hat{c})^2 - \frac{4\delta S_m(\hat{c})}{\beta h_m(\hat{c})}}}{2}. \quad (16)$$

The nonnegative discriminant is the feasibility condition for implementing $\hat{c}$, and the selected branch satisfies $\hat{c} \leq \hat{p}_1^\delta < \hat{p}_2$. Equation (15) links the rescue price to initial coverage through the driver's waiting incentive.

Step 3: optimize the single remaining state variable. Substitute (13) and (16) into completion:

$$J_{\delta,m}(\hat{c}) = [1 - \hat{p}_1^\delta(\hat{c}; m)](1 - e^{-m\hat{c}}) + e^{-m\hat{c}} \frac{S_m(\hat{c})}{\beta}. \quad (17)$$

The first term is initial completion: posting probability times initial coverage. The second is the telescoped continuation term above, multiplied by the probability of initial failure. For $\beta \geq 1/2$, the optimal rescue-menu solution is characterized by

$$\hat{c}^*(m) \in \arg \max_{\hat{c} \in [0, \beta]} J_{\delta,m}(\hat{c}), \quad p_1^*(m) = \hat{p}_1^\delta(\hat{c}^*(m); m), \quad p_2^*(m) = \hat{p}_2(\hat{c}^*(m); m). \quad (18)$$

For $\beta \geq 1/2$, the no-entry global reduction places every optimizer on the strict envelope just derived, so (18) is equivalent to the original two-price problem for every public m . The cutoff is unique within the anonymous symmetric pure-cutoff WPBE class, but the outer argmax need not be. If $\beta < 1/2$, (18) must also be compared with the flat/repeat-only branch $p_1 \geq \beta$.

4 Equilibrium regimes and a reproducible numerical certificate

Rescue value at a fixed thickness. In the uniform no-entry model, let $M_m(p_1, p_2)$ denote the unconditional completion probability in (11), evaluated at the menu's unique equilibrium cutoff value within the maintained cutoff-WPBE class. Holding all other primitives fixed, define

$$\begin{aligned} M_R^*(m) &= \max_{0 \leq p_1 \leq p_2 \leq 1} M_m(p_1, p_2), \\ M_F^*(m) &= \max_{p_1 \in [0, 1]} M_m(p_1, p_1), \\ V(m) &= M_R^*(m) - M_F^*(m). \end{aligned}$$

Here R denotes the best committed rescue menu and F the best single-price policy. Thus $V(m)$ is the increase in optimized equilibrium completion made possible by the rescue option at the same public

thickness m ; $100V(m)$ is that increase in percentage points. It is not the value of increasing m . Since single-price policies are included among rescue menus, $V(m) \geq 0$ by construction.

For a fixed first price, let

$$p_z(m) = \hat{p}_1^\delta(0; m) = \frac{1 - \sqrt{1 - 4\delta S_m(0)/(\beta m)}}{2}.$$

Thus $p_z(m)$ is the largest first price for which the optimized continuation can be implemented at the reject-all boundary $\hat{c} = 0$. The optimal continuation has three regimes.

First price	WPBE cutoff	Rescue price
$0 \leq p_1 \leq p_z(m)$	reject-all boundary $\hat{c} = 0$	$p_2 = \hat{p}_2(0; m)$
$p_z(m) < p_1 < \beta$	unique $\hat{c} \in (0, p_1)$ satisfying $(p_1 - \hat{c})h_m(\hat{c}) = \delta S_m(\hat{c})/[\beta(1 - p_1)]$	$p_2 = \hat{p}_2(\hat{c}; m)$
$\beta \leq p_1 \leq 1$	rescue unused	all $p_2 \geq p_1$ are outcome-equivalent

Exact benchmark as the thin-market limit

Object	Exact $N = 1$ benchmark	Latent-thickness model
Supply information	one incumbent is publicly known	only m is public; $N^I \sim \text{Pois}(m)$ is hidden
Continuation coverage	linear in the cost interval $p_2 - \hat{c}$	$1 - e^{-\alpha m(p_2 - \hat{c})}$ incorporates a random backup pool
Driver incentive	serve now versus discounted waiting	the same tradeoff plus latent assignment competition and a failure posterior
Price solution	radical formulas and one scalar cutoff	Lambert- W rescue price, quadratic p_1 , and one scalar cutoff

The exact benchmark is the thin-market limit of the latent-thickness model. With $\alpha = 1$, $\gamma = 0$, and $m \downarrow 0$, uniformly over the compact feasible price and cutoff sets,

$$1 - e^{-m(p_2 - \hat{c})} = m(p_2 - \hat{c}) + O(m^2), \quad h_m(\hat{c}) = m + O(m^2).$$

Therefore

$$\frac{S_m(\hat{c}, p_2)}{m} \longrightarrow (\beta - p_2)(p_2 - \hat{c}), \quad \frac{h_m(\hat{c})}{m} \longrightarrow 1.$$

Consequently,

$$\hat{p}_2(\hat{c}; m) \longrightarrow \frac{\beta + \hat{c}}{2}, \quad \hat{p}_1^\delta(\hat{c}; m) \longrightarrow \hat{p}_1(\hat{c}), \quad \frac{J_{\delta, m}(\hat{c})}{m} \longrightarrow J_\delta(\hat{c}).$$

The flat objective divided by m converges uniformly to $p_1(1 - p_1)$. Uniform convergence and the maximum theorem permit the maximization to pass through the limit. Equivalently, a Poisson market contains exactly one incumbent with probability $m + o(m)$ and at least two with probability $o(m)$. For $\beta = \delta = .8$, this nesting gives

$$\lim_{m \downarrow 0} \frac{V(m)}{m} = M_1^* - \frac{1}{4} = \frac{461 - 57\sqrt{57}}{3200} = 0.00958107.$$

The numerator follows directly from $(1261 - 57\sqrt{57}) - 800 = 461 - 57\sqrt{57}$ when the flat value $1/4$ is subtracted from M_1^* . Thus the one-driver improvement is exactly the first-order coefficient of the rescue value near an empty market.

Contribution from latent thickness. Latent thickness adds assignment competition in an unobserved driver pool and Bayesian updating after universal rejection. Their interaction with menu commitment and discounted waiting yields the scalar cutoff-WPBE design problem and the value function $V(m)$.

First-order certificate for the outer problem. At an interior solution, put $R_m = \delta S_m / (\beta h_m)$ and suppress $(\hat{c}; m)$ arguments. Primes denote total derivatives with respect to $\hat{c}$, holding m and all other primitives fixed. The envelope theorem for S_m , direct differentiation of h_m , and implicit differentiation of $(1 - p_1)(p_1 - \hat{c}) = R_m$ give

$$\begin{aligned} \frac{S'_m}{S_m} &= -\frac{1}{\beta - \hat{p}_2}, & \frac{h'_m}{h_m} &= \frac{m}{1 - e^{-m\hat{c}}} - \frac{1}{\hat{c}}, \\ R'_m &= R_m \left(\frac{S'_m}{S_m} - \frac{h'_m}{h_m} \right), & (\hat{p}_1^\delta)' &= \frac{1 - \hat{p}_1^\delta + R'_m}{1 + \hat{c} - 2\hat{p}_1^\delta}. \end{aligned} \quad (19)$$

The scalar first-order condition is

$$J'_{\delta,m}(\hat{c}) = -(1 - e^{-m\hat{c}})(\hat{p}_1^\delta)' + m e^{-m\hat{c}}(1 - \hat{p}_1^\delta) + \frac{e^{-m\hat{c}}}{\beta}(S'_m - m S_m) = 0. \quad (20)$$

The numerical procedure solves (12) and (15), brackets and enumerates every root of the scalar condition (20) on $[0, \beta]$, checks the relevant second-order signs, and compares all stationary and boundary values. Rider best responses are then recovered from (9), and driver deviations are checked on both sides of the cutoff. This gives a global one-dimensional WPBE certificate.

Reduced forms. For a flat price $p_1 = p_2$, the equilibrium cutoff is $\hat{c} = p_1$ and the continuation branch is unused. Hence $f_m(p_1) = (1 - p_1)(1 - e^{-mp_1})$, with

$$f'_m(p_1) = -(1 - e^{-mp_1}) + m(1 - p_1)e^{-mp_1}, \quad f''_m(p_1) = -e^{-mp_1}\{2m + m^2(1 - p_1)\} < 0.$$

The unique first-order condition is $e^{mp_1} - 1 = m(1 - p_1)$. Setting $y = 1 + m - mp_1$ gives $ye^y = e^{1+m}$ and therefore

$$M_F^*(m) = \max_{p_1 \in [0,1]} (1 - p_1)(1 - e^{-mp_1}), \quad p_F^*(m) = \frac{1 + m - W_0(e^{1+m})}{m} < \frac{1}{2}. \quad (21)$$

Since $e^{m/2} - 1 > m/2$, $f'_m(1/2) < 0$ and the unique flat optimum lies below $1/2$. For $\beta \geq 1/2$, the rescue-menu optimum reduces to

$$M_R^*(m) = \max_{\hat{c} \in [0, \beta]} J_{\delta,m}(\hat{c}). \quad (22)$$

Strict finite-thickness gain. Take $p_1 = p_F^*(m) < 1/2 \leq \beta$ and a sufficiently small strict increment $p_1 < p_2 < \beta$. Its equilibrium has $0 < \hat{c} < p_1$, and (15)–(17) imply

$$M_m(p_1, p_2) = (1 - p_1)m\phi(m\hat{c}) \left[\hat{c} + \frac{p_1 - \hat{c}}{\delta} \right].$$

The bracket is at least p_1 because $\delta \leq 1$, while ϕ is strictly decreasing and $\hat{c} < p_1$. Thus

$$M_m(p_1, p_2) > (1 - p_1)m\phi(mp_1)p_1 = (1 - p_1)(1 - e^{-mp_1}) = M_F^*(m).$$

This same-first-price comparison proves $V(m) > 0$ at every finite m .

Interior-thickness mechanism. When m is small, there is rarely a driver to rescue. As m grows, universal rejection still occurs but the failed request is increasingly likely to face a useful backup set, so a contingent p_2 creates option value. When m is very large, the optimized flat price already completes almost every request; the remaining improvement must again shrink.

Analytical properties of $V(m)$

Under $F = G = U[0, 1]$, $\gamma = 0$, $\alpha \in (0, 1]$, $\beta \geq 1/2$, and $\delta \in (0, 1]$, the following properties hold for every finite $m > 0$:

$$V(m) > 0, \quad V \text{ is continuous}, \quad \lim_{m \downarrow 0} V(m) = \lim_{m \rightarrow \infty} V(m) = 0.$$

Hence there is some $m^* \in (0, \infty)$ such that

$$V(m^*) = \max_{m > 0} V(m) > 0.$$

This guarantees a finite interior maximizer, but not uniqueness or single-peakedness.

Endpoint argument. On every compact interval of positive m , the continuous extensions at $\hat{c} = 0$ make $J_{\delta, m}(\hat{c})$ jointly continuous in $(m, \hat{c})$. The maximum theorem applied to (21)–(22) makes both optimized values, and hence V , continuous; optimizer uniqueness is unnecessary.

As $m \downarrow 0$, any completion requires at least one incumbent, so

$$0 \leq V(m) \leq M_R^*(m) \leq 1 - e^{-m} \leq m \rightarrow 0.$$

As $m \rightarrow \infty$, the feasible flat price $p_1 = (\log m)/m$ completes with probability $(1 - \log(m)/m)(1 - 1/m) \rightarrow 1$. Therefore $0 \leq V(m) \leq 1 - M_F^*(m) \rightarrow 0$. Positivity at any fixed m_0 , these endpoint bounds, continuity, and the Weierstrass theorem then place at least one global maximizer in a finite interval inside $(0, \infty)$; they do not imply uniqueness or single-peakedness.

Tail rates. For the plotted calibration, the thin-market calculation above gives

$$V(m) \sim \kappa m, \quad \kappa = \frac{461 - 57\sqrt{57}}{3200} = 0.00958107, \quad (m \downarrow 0).$$

For the thick-market rate, put $x = m\hat{c}$ and $T = S_m/\beta$. Equation (17) has the exact loss decomposition

$$1 - J_{\delta, m} = p_1(1 - e^{-x}) + (1 - T)e^{-x}.$$

Using (12)–(16), the feasible choice $x = \log \log m$ yields $p_1 = (1 + o(1))x/m$ and $(1 - T)e^{-x} = O(1/m)$. Conversely, for $p_2 \leq \beta/2$ every feasible policy satisfies

$$m(1 - J_{\delta, m}) \geq x(1 - e^{-x}) + C_m e^{-x}, \quad C_m = \frac{\log(1 + \alpha m \beta / 2)}{\alpha \beta},$$

whose minimum is asymptotic to $\log C_m \sim \log \log m$; policies with $p_2 > \beta/2$ have a larger first loss term. Hence

$$1 - M_R^*(m) \sim \frac{\log \log m}{m}.$$

For the flat optimizer, let $x_F = mp_F^*$. Its first-order condition gives $e^{x_F} = 1 + m - x_F$, so

$$1 - M_F^*(m) = \frac{x_F + 1 - e^{-x_F}}{m} \sim \frac{\log m}{m}.$$

Thus, for fixed $\alpha, \beta > 0$, $\delta \in (0, 1]$, and $\gamma = 0$, $V(m) \sim (\log m)/m$. The displayed single-hump shape below remains numerical evidence rather than an analytical shape result.

5 The value of rescue across market thickness, $V(m)$

All reported values below are deterministic optimization results, not Monte Carlo estimates. For each m , the calculation solves (12) for $p_2(\hat{c})$, uses (16) for $p_1(\hat{c})$, enumerates all roots of (20) together with the boundaries $\hat{c} = 0, \beta$, and retains the candidate with the largest value in (17). It solves (21) for the flat benchmark and then subtracts. The outer search brackets a root of the envelope derivative and checks a wide log grid; the endpoint results above control the two tails. Thus $\hat{m}$ below is the largest value located by this search, not a claim of an analytically unique maximizer.

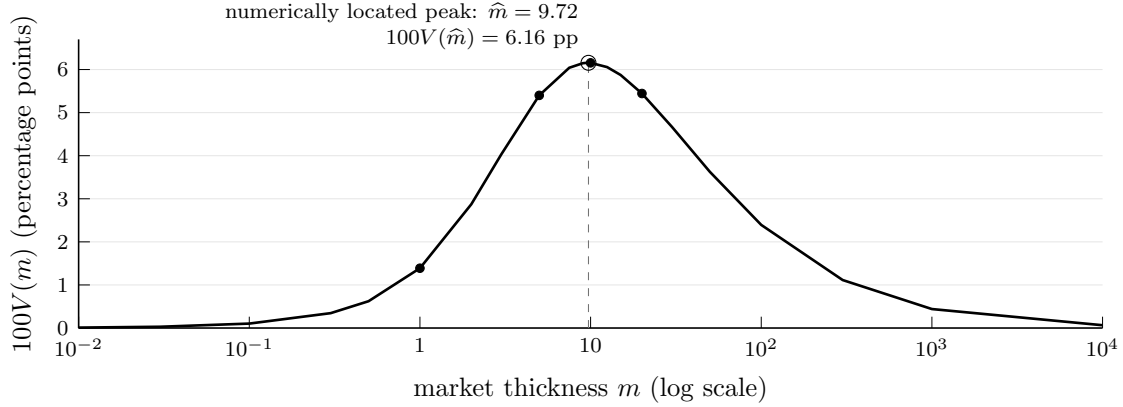


Figure 4: Computed $V(m)$ for $(\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0)$. The solid line connects deterministic optimized values, and black dots mark $m = 1, 5, 10, 20$. Positivity, continuity, and the two endpoint limits are analytical; the displayed hump is numerical evidence.

The table reports maximizers selected by the numerical procedure; their analytical uniqueness is not asserted. Both M_R^* and M_F^* are probabilities on $[0, 1]$; the final column reports their difference in percentage points.

m	$\hat{c}^*(m)$	$p_1^*(m)$	$p_2^*(m)$	$M_R^*(m)$	$M_F^*(m)$	$100V(m)$ (pp)
1	.320340	.391928	.546378	.213230	.199346	1.3883
5	.197789	.236946	.413098	.597813	.543810	5.4003
10	.132863	.155922	.310306	.755154	.693602	6.1552
20	.080430	.092538	.208054	.862392	.807956	5.4437

The largest value located by the outer search occurs at $\hat{m} = 9.718617$, where the selected menu is $(\hat{c}^*, p_1^*, p_2^*) = (.135370, .159002, .314745)$.

Numerical illustration at $m = 10$. The selected menu is $(\hat{c}^*, p_1^*, p_2^*) = (.132863, .155922, .310306)$, while the best flat price is $p_F^* = .217733$. Completion decomposes as follows.

Policy	Complete now	Complete later	Total
Best flat	69.3602%	0	69.3602%
Optimal rescue	62.0532%	13.4622%	75.5154%
Difference	-7.3070 pp	+13.4622 pp	+6.1552 pp

The later column combines repeat and rescue. Conditional on posting, initial failure occurs with probability $e^{-10\hat{c}^*} = 26.4840\%$. Multiplying this failure probability by the continuation completion term in (17) gives 13.4622% unconditional terminal completion, which more than offsets the loss of immediate completion.

Type regions across market thickness

For each row in the table, let

$$C_j^*(m) = 1 - e^{-m[p_j^*(m) - \hat{c}^*(m)]}, \quad v^{M^*}(m) = \frac{p_2^*(m)C_2^*(m) - p_1^*(m)C_1^*(m)}{\beta[C_2^*(m) - C_1^*(m)]}.$$

The rider widths are $(p_1^*, p_1^*/\beta - p_1^*, v^{M^*} - p_1^*/\beta, 1 - v^{M^*})$; the driver widths are $(\hat{c}^*, p_1^* - \hat{c}^*, p_2^* - p_1^*, 1 - p_2^*)$. They have the same four meanings as in Figure 3, but both the optimal menu and the rider's coverage-adjusted switching value are recomputed at every m .

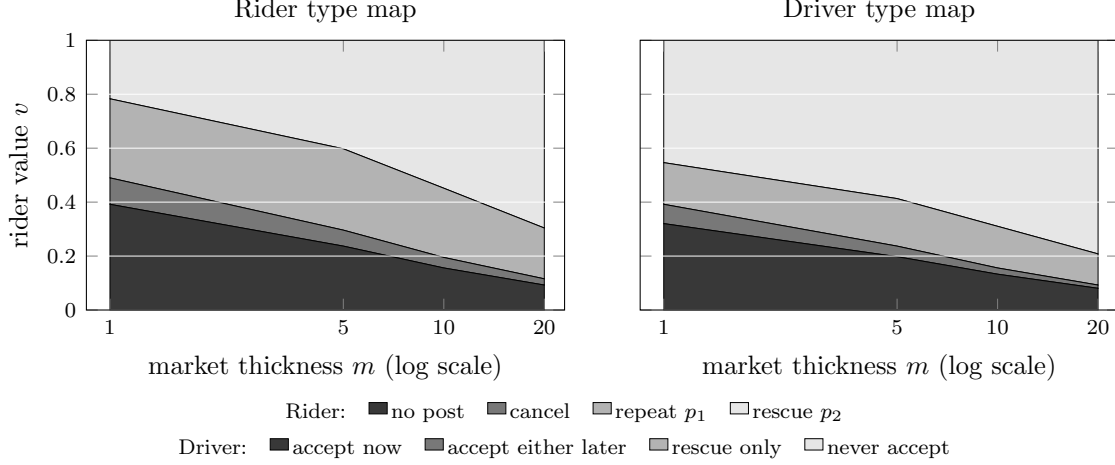


Figure 5: Equilibrium type maps at $m = 1, 5, 10, 20$ for $(\alpha, \beta, \delta, \gamma) = (1, .8, .8, 0)$. Each vertical slice partitions $[0, 1]$ and the logarithmic horizontal scale separates the four recomputed markets; connecting segments are visual guides. The bands are normalized type shares, not realized path probabilities; expected incumbent counts equal m times the driver shares. The rider rescue region grows from .2169 to .6960, while pool-level rescue coverage $C_2^*(m)$ rises from .2023 to .9221.

Claim boundary. Strict single-peakedness, uniqueness of the maximizing m , and monotonicity or uniqueness of optimal menus remain open; this calibration is consistent with them. Global design with fresh entry $\gamma > 0$ is also open.